Optimizing Financial Portfolios: A Mathematical Approach Using Covariance Matrices and Linear Algebra

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Abstract

This project investigates the application of mathematical modeling to optimize financial portfolios—specifically, portfolios composed of major U.S. banking institutions, including JPMorgan Chase, Bank of America, and Wells Fargo. Portfolio optimization involves allocating assets to maximize returns while minimizing risk, a central problem in quantitative finance that draws on tools from linear algebra and statistics. The study applies Markowitz's Modern Portfolio Theory (MPT) to estimate mean returns, standard deviations, and covariances among assets, constructing the covariance matrix that underpins the optimization process.

Using historical stock price data from publicly available sources such as Yahoo Finance, the research applies a mean–variance optimization approach to identify the minimum-variance portfolio and visualize the efficient frontier, illustrating the trade-off between risk and return. This framework provides a mathematically rigorous and practically interpretable method for managing financial risk. The results present optimal asset allocations and a visual representation of portfolio efficiency, demonstrating how linear algebra and statistical modeling yield transparent and data-driven insights for investment decision-making

I. Introduction & Background

Portfolio optimization is a cornerstone of modern finance, shaping investment strategies and guiding effective risk-management practices (Ahmed, Saeed, & Khan, 2022). In today's increasingly data-driven and volatile markets, mathematical modeling provides a systematic framework for efficient asset allocation by quantifying the relationship between risk and return (Broby, 2022). This study investigates the optimization of banking portfolios—specifically those composed of major U.S. financial institutions such as JPMorgan Chase, Bank of America, and Wells Fargo—to illustrate how statistical and algebraic methods can be applied to real-world financial data.

The analysis focuses on key attributes of these portfolios, including mean return, standard deviation, covariance, and correlation of asset returns. These quantitative measures form the statistical foundation for the Markowitz Mean–Variance Portfolio (MVP) Model, which balances expected return against portfolio risk (Markowitz, 1952). Balancing risk and return is essential because rational investors seek to maximize returns without taking on unnecessary volatility. In practice, this involves determining the combination of assets whose interaction minimizes overall portfolio variance while maintaining a desired expected return (Engels, 2004).

This objective is achieved by constructing covariance matrices to model relationships between asset returns and applying linear-algebraic techniques to identify the optimal asset weights that minimize portfolio variance for a specified level of return (Ahmed et al., 2022). The resulting allocations are visualized on the efficient frontier, which represents all optimal portfolios under varying risk preferences. The visualization not only demonstrates the mathematical structure underlying portfolio selection but also provides an interpretive tool for risk management in financial institutions (Royal Society, 2020).

Ultimately, this project underscores the interdisciplinary significance of combining linear algebra, statistics, and finance to solve practical decision-making problems. Markowitz's Modern Portfolio Theory (MPT) remains central to this discipline because it introduces a rigorous quantitative approach to investment strategy that aligns mathematical optimization with economic rationality (Markowitz, 1952; Engels, 2004). This methodological framework demonstrates how mathematical models serve as a bridge between theoretical precision and practical financial relevance—linking academic insight with the real-world goals of investors, analysts, and portfolio managers.

II. Background Literature Review

Our project on portfolio optimization within the banking sector builds upon a rich body of theoretical and applied research in quantitative finance. The story of modern portfolio management truly begins with Harry Markowitz's seminal work on Modern Portfolio Theory (1952). Markowitz revolutionized financial decision-making by demonstrating that investors need not accept risk passively; rather, they can strategically allocate assets to minimize overall risk for a desired level of return. His insight lay in recognizing that an asset should not be evaluated in isolation—its relationship with every other asset matters. This interdependence is captured mathematically through covariance matrices, which quantify the tendency of assets to move together within a portfolio (Markowitz, 1952; Engels, 2004).

Covariance matrices are more than abstract mathematical constructs; they are the empirical engines of portfolio diversification. They allow financial analysts and institutions to measure how asset pairs co-vary—whether the returns of two securities rise and fall together (high covariance) or move in opposite directions (negative covariance). In the banking environment, where portfolios often combine equities, bonds, and derivative instruments, these correlations directly inform risk exposure and regulatory capital allocation. As explained by Ahmed, Saeed, and Khan (2022), effective portfolio optimization in risk management depends on accurate covariance estimation, since it serves as the statistical core for determining overall portfolio volatility. Their work validates the use of covariance matrices and mean-variance models to identify optimal asset mixes, a methodology closely mirrored in our study focused on minimizing variance within a three-bank portfolio—JPMorgan Chase, Bank of America, and Wells Fargo. Our project extends their findings by applying these concepts specifically to the U.S. banking sector, using real market data to construct an operational framework for minimum-risk portfolio selection.

Implementing the principles of MPT and covariance analysis requires robust computational and mathematical tools—chief among them, linear algebra. As noted by Broby (2022), linear algebra underpins modern financial analytics by providing the mathematical structure to solve large-scale optimization problems subject to constraints. Banking institutions routinely handle complex products and massive datasets, making efficient computation of risk and return relationships essential. The use of matrix operations, vector representations, and linear systems is fundamental for calculating portfolio variance and covariances, as well as for solving the optimization equations that produce the Minimum Variance Portfolio (MVP). In this project, linear-algebraic techniques are employed to invert covariance matrices and compute optimal weight vectors—key steps in determining how capital should be allocated across assets to achieve the lowest possible portfolio risk (Ahmed et al., 2022; Markowitz, 1952). Such optimization directly supports risk-management objectives in the banking sector, where capital-adequacy regulations require institutions to balance risk exposure with return efficiency (Royal Society, 2020).

While this existing literature—spanning Markowitz's theoretical foundation, the statistical modeling of covariance, and computational approaches through linear algebra—provides the conceptual backbone of modern portfolio theory, our research contributes a practical bridge between these methods and their application to real-world bank portfolios. Specifically, we translate these mathematical concepts into a computational modeling framework that calculates monthly returns, derives covariance matrices, and identifies the

Minimum Variance Portfolio using actual historical stock data from major U.S. banks. Through this application, we move from theory to implementation, illustrating how tools of linear algebra and statistics can be operationalized to minimize risk in active financial decision-making. This hands-on approach distinguishes our work by showing how Markowitz's model—often discussed in abstract mathematical terms—can be repurposed as an accessible, data-driven risk-optimization tool for the modern banking landscape.

III. **Theoretical Framework**

Definitions and Notation

Portfolio construction involves selecting a collection of n assets, each assigned a specific weight, denoted as w_i (Markowitz, 1952). The expected return of a portfolio, represented as μ_p, is calculated as the weighted average of the individual asset returns: $\mu_p = \Sigma_{i=1}^{n} w_i\mu_i$ (Broby, 2022). Portfolio variance, symbolized as σ_p^2, serves as a measure of risk and is computed using the formula $\sigma_p^2 = w^T\Sigma w$, where w represents the weight vector and Σ signifies the covariance matrix of asset returns ("Portfolio Optimization in Risk Management", 2022). The foundation of portfolio optimization rests on key financial theorems. The Efficient Frontier Theorem posits that for a given set of assets, the efficient frontier delineates the set of portfolios offering the highest possible return for any given level of risk (Markowitz, 1952). This leads to a core optimization problem: to minimize portfolio variance (σ_p^2) subject to specific constraints, such as the sum of asset weights equaling one ($\Sigma w_i = 1$) (Broby, 2022).This problem is mathematically framed as a constrained quadratic optimization problem: Minimize:$w^T\Sigma w$ Subject to:$\Sigma w_i = 1$Here, w is the vector of asset weights and Σ is the covariance matrix of asset returns (Ahmed et al., 2022). Solving this requires determining the optimal weights w through methods like Lagrange multipliers or numerical optimization techniques ("The Role of Linear Algebra in Financial Analytics", 2022).

IV. **Methodology**

This study examines nine U.S. banking institutions, organized into three performance-based groups:

(1) three successful large banks (e.g., JPMorgan Chase, Bank of America, Wells Fargo),

(2) three mid-sized banks with moderate stability, and

(3) three failing or recently closed banks.

This classification supports comparative analysis of portfolio optimization across distinct financial health categories. Historical monthly adjusted closing prices for all nine banks are collected from Yahoo Finance for the period January 2024 through November 2024 using the Python `yfinance` library.

After computing MVPs for each category, returns from all nine banks are combined to estimate a comprehensive Minimum Variance Portfolio (MVP) representing the U.S. banking sector collectively. Focusing on the 2024 period provides a snapshot of modern banking performance under present economic conditions, particularly within a year characterized by fluctuating interest rates and evolving regulatory contexts. These conditions make the data especially relevant for analyzing portfolio risk optimization in a contemporary setting. Each asset's dataset includes:

- Date (month end);
- Adjusted closing price (USD); and
- Monthly return (%).

Adjusted prices are chosen because they reflect the equity's true market performance, correcting for corporate actions such as dividends and stock splits (Brogaard & Detzel, 2015). Data is sourced programmatically using the `yfinance` library in Python. From each stock's price series, monthly simple returns are computed as the percentage change from one month's adjusted closing price to the next:

$$Rt = Pt - Pt\text{-}1/Pt\text{-}1$$

(Elton, Gruber, Brown, & Goetzmann, 2009). This transformation converts price data into measures of performance that are comparable across assets. Using monthly frequency helps smooth short-term volatility and better reflects medium-term portfolio behavior (Broby, 2022). The resulting variables include:

- Expected monthly return (mean of all monthly returns),
- Volatility (standard deviation of monthly returns), and
- Covariance matrix (Σ), capturing how returns move together.

These quantities provide the empirical input for implementing Modern Portfolio Theory (MPT) in identifying diversification benefits and optimal portfolio allocations (Markowitz, 1952; Ahmed, Saeed, & Khan, 2022). Expected returns E(R_i) are computed as the average of each asset's monthly returns over January–November 2024:

$$E\,R_i = \frac{1}{n} \sum_{t=1}^{1\;N} R_{i,t}$$

The covariance matrix (Σ), representing paired co-movements between all five asset returns, is calculated in Python using NumPy: Σ = cov(R) (Broby, 2022).

These computations establish the data needed to quantify portfolio risk and diversification. The optimization process seeks the set of weights w that minimizes portfolio variance subject to full investment (Markowitz, 1952):

Minimize: $w^T \Sigma w$

Subject to:

$$\Sigma_{i\text{-}1\text{-}1} \; {}_{m} \; w_i = 1$$

Here, wi is the share of total capital allocated to asset i. Portfolio expected return and standard deviation are computed as: $\mu p = w^\wedge T\mu$, $\sigma p = \sqrt{}w^\wedge T\Sigma w$

Optimization is performed using Python's SciPy library (`optimize.minimize`), which efficiently solves constrained quadratic problems (Chong & Loong, 2013). Solutions are then plotted to form the Efficient Frontier, which visualizes all Pareto-optimal risk-return combinations. The Minimum Variance Portfolio (MVP) appears at the frontier's leftmost point, reflecting the portfolio with the lowest total risk for 2024's data (Broby, 2022).

The Efficient Frontier is visualized with Matplotlib and Seaborn, displaying risk on the x-axis and expected return on the y-axis. Individual assets appear as points below or on the curve, demonstrating how diversification improves outcomes (Elton et al., 2009).

The process involves:

1. Retrieving January–November 2024 data using `yfinance`.
2. Generating monthly returns via `pandas`.
3. Building covariance and mean-return matrices with `numpy`.
4. Running optimization with `scipy`.
5. Plotting results clearly and labeling the MVP.

All code was executed in Python 3.10. Data summaries and selected output tables were exported to Excel for verification and reporting (McKinney, 2018).

What Is Portfolio Optimization and What Makes a Portfolio Optimal?

At its core, portfolio optimization is about deciding how to allocate wealth across different assets—such as stocks, bonds, or other investments—in order to achieve the best possible balance between risk and return. Every investment has an expected payoff but also a degree of uncertainty, and that uncertainty represents risk. The goal of portfolio optimization is to choose the combination of assets that provides the highest expected return for a given level of risk, or equivalently, the lowest risk for a desired level of expected return.

This trade-off lies at the heart of modern finance. Investors—especially those who are risk averse—want to avoid large losses even if it means giving up some potential profit. As Engels (2004) explains through a simple casino example, people often prefer outcomes with smaller but more predictable returns over those with higher potential gains that carry a much larger chance of loss. In finance, this behavior translates into constructing a portfolio that avoids unnecessary volatility while still generating strong, consistent returns.

Portfolio optimization became formally defined with the work of Harry Markowitz (1952), whose mean–variance analysis introduced a mathematical framework for quantifying both return and risk. Markowitz showed that an investor should not evaluate assets in isolation but instead look at how they interact within a portfolio. Two investments that both seem risky individually can, when combined, yield a portfolio with lower overall risk if their price movements are not perfectly correlated. This insight led to the concept of the efficient frontier —the set of all “optimal” portfolios that offer the maximum expected return for each level of risk (Markowitz, 1952).

An optimal portfolio, therefore, is one that lies on this efficient frontier. At that point, the investor cannot improve the expected return without taking on additional risk, nor can they reduce risk without sacrificing return. In practical terms, portfolio optimization helps investors—ranging from individuals to large financial institutions—allocate capital strategically, guard against downside volatility, and achieve objectives that align with their risk tolerance and investment goals. It turns subjective preferences about risk into a systematic, data-driven process grounded in statistical analysis of asset returns and covariances, which are precisely the data inputs used in this research.

Introduction to the Markowitz Model: What is it?

The foundation of modern portfolio optimization began with the work of Harry Markowitz, an American economist who transformed investment theory through his 1952 paper "Portfolio Selection" published in The Journal of Finance. In this seminal work, Markowitz introduced what became known as Modern Portfolio Theory (MPT)—a framework for constructing portfolios that seek to maximize expected return for a given level of risk, or equivalently, minimize risk for a given expected return (Markowitz, 1952). For this breakthrough, he was later awarded the Nobel Prize in Economic Sciences in 1990.

Prior to Markowitz, investment selection tended to evaluate securities independently, focusing mainly on their individual returns. Markowitz's critical insight was that the total risk of a portfolio depends not only on each asset's volatility but also on how asset returns move together, captured mathematically through covariances. This led to the development of the efficient frontier, which represents the set of optimal portfolios offering the best possible return for each level of risk (Elton et al., 2014).

In practice, these expected returns and covariances are estimated from historical data using the rates of return for each asset—typically calculated as the percentage change or logarithmic change in price over time. These observed rates of return provide the empirical basis of the Markowitz model: they form the inputs for both the expected return vector and the covariance matrix that underpin portfolio optimization. Accordingly, the mathematical problem is expressed as minimizing portfolio variance, $\sigma_p^2 = w^T \Sigma w$ subject to constraints on portfolio weights w and target return μp = w^Tμ, where

∑ is the covariance matrix of historical returns and μ is the vector of mean (expected) returns.

The Markowitz model remains one of the most influential and actively applied frameworks in finance. It provides the foundation for modern approaches to asset allocation, risk management, and investment performance evaluation. In this research, the model serves as the theoretical framework connecting the calculated rates of return to optimal portfolio selection and analysis of risk–return trade-offs.

What exactly are we doing? What approach are we taking?

So we're gonna run through four groups of banks. For this research we're going to look at three data sets of banks, one that has three very large and successful banks, one that has three regular or potentially even small banks, and one that has three failing or closed banks. And then at the end of all of that we are going to combine the stats of all the groups together and get an MVP for all of the banks together. As stated earlier we're going to look through databases in yahoo finance and calculate the rate of returns, then we'll use the data we get from each banks database and calculate what we need for the MVP model, and then we will plug those values and in and find out our MVP for each group of banks So the first bank we are looking at is JPMorgan. This is one of the biggest and most successful banks in America. In order for us to calculate our MVP, first we need the rate of return for these banks. As I mentioned, we will be using three groups, each with three banks, and I will calculate the rate of return between each month and the adjusting closing prices, and then the stats I get from all three banks will be used to calculate the values we need for our MVP model.

Successful Bank Portfolio Optimization

This section constructs and analyzes a minimum-variance portfolio (MVP) using three of the most established U.S. banks frequently regarded as successful industry leaders:

- JPMorgan Chase (JPM)
- Bank of America (BOA)

- Wells Fargo (WFC)

All results are based on monthly returns from January 2021 through November 2025.

The goal is to determine the portfolio combination offering the lowest overall risk while maintaining exposure to this subset of top-performing banks.

Monthly Returns Dataset: Successful Banks (JPMorgan, Bank of America, Wells Fargo)

A shortened view of the dataset is presented below (full series in appendix):

Date	JPM	BOA	WFC
Nov 2025	−0.0194	−0.0157	−0.0221
Oct 2025	−0.0137	0.0419	0.0377
Sep 2025	0.0465	0.0166	0.0260
Aug 2025	0.0224	0.0734	0.0192
Jul 2025	0.0218	0.0049	0.0063
...	...	...	...
Jan 2021	0.0126	−0.0157	−0.0100

This is shown by calculating the returns based on the adjusting closing prices from the bank data we got from Yahoo Finance. The formula for returns is Rt = Pt - Pt-1/Pt-1. What this essentially means is when we look at the data in Yahoo Finance, we are looking at the starting date of one month then subtracting that adjusted closing price by the prior month, and then dividing that by the prior months closing price. For example, if we look at the data in Yahoo Finance for adjusting closing price, we see that Sep 1-25 has an adjusted closing price of $313.90, where as the month prior Aug 1-25 has a price of $299.96. With this information our formula would look like this: 313.90 – 299.96/299.96 = 0.0465. This is what we get for Sep-25 Rate of Return. I'm not going to show all of these calculation because that would be long and tedious, not to mention boring, but now you know how the rate of returns were calculated here.

Descriptive Statistics

Bank	Mean Return	Std. Deviation
JPM	0.01950	0.07043
BOA	0.01479	0.08229
WFC	0.02294	0.08259

Finding this is just basic statistics, we find the mean by adding up all the return values we got for each bank, we add them altogether, and then divide by the number of values where in this case we have 58 values so we divide that big sum by 58 and that's how we get our mean for each bank. Standard deviation has a more complex formula that almost no one uses because its long, complicated, and in many ways impractical. I just used the function in excel "STDEV.S" and then highlighted all the returns we had for each bank and calculated the standard deviation for each bank.

These results show that Wells Fargo (WFC) had the highest average monthly return (2.29 %) and slightly higher risk, while JPMorgan Chase (JPM) returned a nearly comparable average with the lowest volatility of the three.

Step 1: Covariance Matrix

	JPM	BOA	WFC
JPM	0.004877	0.004919	0.004373
BOA	0.004919	0.006657	0.005874
WFC	0.004373	0.005874	0.006705

To see how the banks move together — sometimes up, sometimes in opposite directions — we create a covariance matrix. This tells us how much the returns of two banks tend to change in the same direction.

These are monthly covariances (the units are "squared returns").

A positive number means the pair tends to move in the same direction.

This is another example like standard deviation where the formula is more complex, I just did the excel function "Covariance.P," except in this case its not as simple as going through all the values of each individual bank and calculating it for each, Covariance requires two sets of data. In this case we're going to highlight all of the returns for JPM and then highlight it again for our second set of data, so we get covariance of JPM with just its own data going into itself, I just call this 1:1. Then we do that for BOA and WFC which gets us 2:2 and 3:3, and then we combine the data sets in covariance like taking JPM's returns and putting it together with BOA and WFC's data in the covariance formula, there we get 1:2 and 1:3, and then BOA and WFC go into covariance formula which gets us 2:3. When we make the matrix we start with our first row with three values, those three values will be whatever we get for 1:1, 1:2, and 1:3. Then we go to the second row and do 2:1, 2:2, and 2:3, and same for the third row. We only have three banks so this will be a 3x3 matrix. Notice also how some of the values repeat, that's because some these values are essentially the same calculations, like 1:3 is the same as 3:1 because its using the same sets of data just in a different order. The only unique values are the symmetric ones like 1:1, 2:2, and 3:3. All covariances are positive, indicating the three major banks tend to move together. Despite this, their differences in magnitude still allow some diversification benefit.

Step 2: Finding the Minimum-Variance Portfolio

The goal is to find the mix of the three banks (their weights) that gives the portfolio the lowest possible risk. Mathematically, this involves computing:

$$\mathbf{w} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^{\top}\Sigma^{-1}\mathbf{1}}$$

where

- Σ^-1 = the inverted covariance matrix

- 1 = a column of ones (representing each stock)

Step 3: Inverted Covariance Matrix

	JPM	BOA	WFC
JPM	805.759	–580.301	–17.090
BOA	–580.301	1079.731	–567.477
WFC	–17.090	–567.477	657.430

This inverted matrix is used to derive the weights that minimize portfolio variance.

We got these values by using the "Minverse" function on excel with our covariance matrix that we just made and it gives us this inverted matrix

Step 4: Matrix Multiplication with a Vector of Ones

Multiply the inverted covariance matrix by a column vector of 1's:

This part requires matrix multiplication. By this point we should also have a 1x3 matrix that all just says 1 because these values will end up equaling 1. We have a 3x3 inverted matrix, and 1x3 1 matrix, how this multiplication works is we multiply the three values in our first row of our 3x3 matrix and multiply it downwards from the 1 matrix. After we get those three values we add them together, and that gets us our first out of three values we should end up getting for this matrix. In this case the first value will look like this (805.759)(1) + (-580.301)(1) + (-17.090)(1) = 208.368. This gets us our first value for our new 1x3 matrix, and using the same process we should get 208.368, -68.048, and 72.863. These are the unnormalized weight components.

Step 5: Normalizing Weights

Normalize each value by dividing by the total sum of all three:

Sum = 208.368 - 68.048 + 72.863 = 213.183

Each weight = that value ÷ 213.183

When we've done this, this essentially translated into what our denominator for the weights formula is. With this we would do 208.368/213.183, -68.048/213.183, and 72.863/213.183, which get us our new weights of 0.977413223, -0.319199669, and 0.341786446, and all of these add up to 1 as well which satisfies the portfolio condition. These will be used to calculate our expected return and our portfolio variance and standard deviation

Weights are therefore:

Bank	Raw Value	Normalized Weight
JPM	208.368	**0.9774**
BOA	–68.048	**–0.3192**
WFC	72.863	**0.3418**

Step 6: Interpretation of Weights

- JPM (0.9774) → Primary holding in the portfolio
- BOA (–0.3192) → Short or hedge allocation reducing portfolio volatility
- WFC (0.3418) → Secondary contributor supporting return potential

This combination mathematically produces the minimum achievable risk for this set of assets.

Step 7: Expected Portfolio Return

Expected Portfolio Return asks for us to use dot product. We're doing dot product with our weight vector (0.977413223, -0.319199669, 0.341786446), and our mean or average vector (0.019504836, 0.014793682, 0.022939803). In this case when we do our multiplication and add everything together, we should end up getting 0.02218266 or 2.22%

Expected Monthly Return = 2.22 %

Step 8: Portfolio Variance and Standard Deviation

Next is to calculate the Portfolio Variance (Risk). This was a little more complicated so I just used excel to find this, and the formula looked like this

=MMULT(TRANSPOSE(weights), MMULT(covariance_matrix, weights))

My result was 0.004690804

Then in order to find our portfolio standard deviation we just take the variance that we just found and square root it which ends up getting 0.068489448 or 6.85%

Portfolio Standard Deviation = 6.85 % per month

Step 9: Sharpe Ratio

To compare returns relative to risk, we calculate the Sharpe Ratio.

Formula:

S = Expected returns – risk free rate/portfolio standard deviation

We already calculated expected returns and portfolio standard deviation which was 0.02218266 and 0.068489448, but what is risk free rate? This is a rate given at certain times to banks. As of November 14, 2025, the 3-month treasury bill yield (risk free rate) was 3.95%, and since this is in a year, there would be approximately 0.33% monthly. From this our formula would look like

S = 0.02218266 – 0.0033/0.068489448 = -0.158

Sharpe Ratio = -0.158 per month, or about -0.55 annualized

$(0.105 \times \sqrt{12})$

Interpretation:

For every 1 unit of risk, this portfolio earns about -0.158 units of return above the risk-free rate.

Interpretation

- The Sharpe ratio is moderately negative with this high assumed risk-free rate.
- The portfolio still earns positive returns but does not exceed the benchmark return after adjusting for risk.

- Relative to the earlier groups, this portfolio delivers higher raw return and lower volatility, but when compared against an unusually high monthly risk-free rate, it underperforms in excess-return terms.

Summary of Results

Weights (JPM, BOA, WFC): [0.9774, –0.3192, 0.3418]

Expected Monthly Return: 0.02218 (2.22 %)

Portfolio Variance: 0.0046908

Portfolio Std. Dev.: 0.06849 (6.85 %)

Sharpe Ratio: –0.158

Conclusion

Through this exercise, we demonstrated how to systematically:

1. Measure the average returns and risks of individual major banks.

2. Examine their interrelationships with covariance analysis.

3. Use matrix algebra (or Excel functions) to identify the allocation that minimizes total portfolio variance.

4. Assess performance relative to the risk-free rate using the Sharpe ratio.

The minimum-variance portfolio for JPM, BOA, and WFC achieved an expected monthly return of 2.22 % with a risk level of 6.85 %, producing a Sharpe ratio of –0.158 when evaluated against a 3.3 % monthly risk-free rate.

This outcome shows that while these leading banks provide strong absolute stability and growth relative to smaller or regional banks, the high benchmark risk-free rate causes their risk-adjusted performance to appear negative. Nevertheless, the portfolio reflects how combining industry leaders can deliver greater consistency and efficiency—a practical lesson in how diversification optimizes even top-tier assets.

Regular Bank Portfolio Optimization

This section demonstrates the step-by-step calculation of a minimum-variance portfolio (MVP) using three large, regionally focused U.S. banks:

- Fifth Third Bancorp (FITB)

- Comerica Bank (CMA)

- Truist Financial Corporation (TFC)

All calculations are based on monthly returns from

January 2021 through November 2025. The goal is to determine the combination of these three securities that produces the lowest possible portfolio variance, given their historical risk relationships.

Monthly Returns Dataset: Regular Banks (Fifth Third Bancorp, Comerica Bank, Truist Financial Corporation)

The dataset includes monthly percentage returns (in decimal format).

A shortened section of the table is presented below; the full version would appear in an appendix.

Date	FITB	CMA	TFC
Nov 2025	0.0300	0.0312	0.0025
Oct 2025	–0.0575	0.1278	–0.0238
Sep 2025	–0.0267	–0.0292	–0.0115
Aug 2025	0.1012	0.0446	0.0711
Jul 2025	0.0198	0.1469	0.0167
...	...	...	...
Jan 2021	0.0599	0.0374	0.0010

This is shown by calculating the returns based on the adjusting closing prices from the bank data we got from Yahoo Finance. The formula for returns is Rt = Pt - Pt-1/Pt-1.

What this essentially means is when we look at the data in Yahoo Finance, we are looking at the starting date of one month then subtracting that adjusted closing price by the prior month, and then dividing that by the prior months closing price. For example, if we look at the data in Yahoo Finance for adjusting closing price, we see that Nov 1-25 has an adjusted closing price of $42.87, where as the month prior Oct 1-25 has a price of $41.62. With this information our formula would look like this: 42.87 – 41.62/41.62 = 0.0300. This is what we get for Nov-25 Rate of Return. I'm not going to show all of these calculation because that would be long and tedious, not to mention boring, but now you know how the rate of returns were calculated here. Each value here shows the return for that month as a decimal (for example, 0.03 = 3 %).

Step 1: Descriptive Statistics

Bank	Mean	Std. Dev.
FITB	0.0148	0.0907
CMA	0.0160	0.1085
TFC	0.0060	0.0820

Finding this is just basic statistics, we find the mean by adding up all the return values we got for each bank, we add them altogether, and then divide by the number of values where in this case we have 58 values so we divide that big sum by 58 and that's how we get our mean for each bank. Standard deviation has a more complex formula that almost no one uses because its long, complicated, and in many ways impractical. I just used the function in excel "STDEV.S" and then highlighted all the returns we had for each bank and calculated the standard deviation for each bank.

- These means represent the historical average monthly growth rate for each stock.
- CMA has the highest average return, while TFC is the least volatile.

Step 2: Covariance Matrix

The covariance matrix (Σ) measures how the banks' returns move relative to each other.

	FITB	CMA	TFC
FITB	0.008091	0.008159	0.006516
CMA	0.008159	0.011582	0.007252
TFC	0.006516	0.007252	0.006605

All entries are positive, which confirms that these three stocks tend to move in similar directions, though not perfectly, which creates diversification potential.

This is another example like standard deviation where the formula is more complex, I just did the excel function "Covariance.P," except in this case its not as simple as going through all the values of each individual bank and calculating it for each, Covariance requires two sets of data. In this case we're going to highlight all of the returns for FITB and then highlight it again for our second set of data, so we get covariance of FITB with just its own data going into itself, I just call this 1:1. Then we do that for CMA and TFC which gets us 2:2 and 3:3, and then we combine the data sets in covariance like taking

FITB's returns and putting it together with CMA and TFC's data I the covariance formula, there we get 1:2 and 1:3, and then CMA and TFC go into covariance formula which gets us 2:3. When we make the matrix we start with our first row with three values, those three values will be whatever we get for 1:1, 1:2, and 1:3. Then we go to the second row and do 2:1, 2:2, and 2:3, and same for the third row. We only have three banks so this will be a 3x3 matrix. Notice also how some of the values repeat, that's because some these values are essentially the same calculations, like 1:3 is the same as 3:1 because its using the same sets of data just in a different order. The only unique values are the symmetric ones like 1:1, 2:2, and 3:3

Step 3: Inverted Covariance Matrix

To solve for the MVP weights, the inverse of the covariance matrix (Σ^{-1}) is calculated as follows:

	FITB	CMA	TFC
FITB	723.168	−200.558	−493.309
CMA	−200.558	331.814	−166.432
TFC	−493.309	−166.432	820.861

We got these values by using the "Minverse" function on excel with our covariance matrix that we just made and it gives us this inverted matrix

Step 4: Multiply by a Column of Ones

This part requires matrix multiplication. By this point we should also have a 1x3 matrix that all just says 1 because these values will end up equaling 1. We have a 3x3 inverted matrix, and 1x3 1 matrix, how this multiplication works is we multiply the three values in our first row of our 3x3 matrix and multiply it downwards from the 1 matrix. After we get those three values we add them together, and that gets us our first out of three values we should end up getting for this matrix. In this case the first value will look like this (723.168)(1) + (-200.558)(1) + (-493.309)(1) = 29.301. This gets us our first value for our new 1x3 matrix, and using the same process we should get -29.301, -35.176, and 161.120 These are the unnormalized weight components.

Step 5: Normalize to Obtain Portfolio Weights

To find usable portfolio weights that sum to 1, divide each element by the total of the three.

Sum = 29.301 – 35.176 + 161.120 = 155.245

With this we would do 29.301/155.245, -35.176/155.245, and 161.120/155.245, which get us our new weights of 0.18874128, -0.2265867, and 1.03784545, and all of these add up to 1 as well which satisfies the portfolio condition. These will be used to calculate our expected return and our portfolio variance and standard deviation

Bank	Raw Value	Normalized Weight
FITB	29.301	0.1887
CMA	–35.176	–0.2266
TFC	161.120	1.0378

Step 6: Interpretation of Weights

- A positive weight (FITB = 0.19 ; TFC = 1.04) means the stock is held long in the portfolio.
- The negative weight on CMA (–0.23) represents a short position, which theoretically earns a positive contribution when CMA's price falls.
- This negative exposure is a mathematical hedge, not necessarily a real short sale, its used to reduce total volatility.

Step 7: Expected Portfolio Return

To estimate portfolio return, multiply each weight by its mean return and sum the results (the dot product):

Expected Portfolio Return asks for the use of the dot product. In this we're doing multiplication with our weight vector (0.18874128, -0.2265867, 1.03784545), and our mean or average vector (0.014798036, 0.01596833, 0.00597478). What we do here is multiply the first value in the weight vector by the first value in the mean vector, then do that process for the second and third, then add them together. The dot product should look something like this: (0.18874128)(0.014798036) + (-0.2265867)(0.01596833) + (1.03784545)(0.00597478). In this case when we do our multiplication and add everything together, we should end up getting 0.00537569 or 0.54%

Expected Monthly Return = 0.00538 (≈ 0.54 %)

This is the combined expected performance of the three-bank portfolio.

Step 8: Portfolio Variance and Volatility

Next is to calculate the Portfolio Variance (Risk). This was a little more complicated so I just used excel to find this, and the formula looked like this

=MMULT(TRANSPOSE(weights), MMULT(covariance matrix, weights))

My result was 0.006441451

Then in order to find our portfolio standard deviation (volatility) we just take the variance that we just found and square root it which ends up getting 0.080258652 or 8.03%.

Portfolio Standard Deviation = 8.03 % per month

Step 9: Sharpe Ratio – Risk-Adjusted Performance

To compare returns relative to risk, we calculate the Sharpe Ratio.

Formula:

S = Expected returns – risk free rate/portfolio standard deviation

We already calculated expected returns and portfolio standard deviation which was 0.00537569 and 0.080258652, but what is risk free rate? This is a rate given at certain times to banks. As of November 14, 2025, the 3-month treasury bill yield (risk free rate) was 3.95%, and since this is in a year, there would be approximately 0.33% monthly. From this our formula would look like

S = 0.00537569 – 0.0033/0.080258652 = -0.344

Sharpe Ratio = -0.344 per month, or about -1.19 annualized

($-0.344 \times \sqrt{12}$)

Interpretation:

For every 1 unit of risk, this portfolio earns about -0.344 units of return above the risk-free rate.

Discussion

The final weights demonstrate how the optimization model balances each bank's contribution to total variance:

- FITB provides moderate stability and return.
- CMA carries a negative weight, functioning as a mathematical offset to reduce portfolio risk.
- TFC dominates the position, reflecting its lower individual variance and covariance characteristics.

Although the expected return is relatively modest, the resulting portfolio achieves the lowest attainable volatility for this set of assets, fulfilling the definition of a minimum-variance portfolio.

Conclusion

Through this exercise, we demonstrated how to systematically:

1. Measure the individual returns and risks of the regular banking stocks (FITB, CMA, TFC).
2. Quantify how their movements relate through the covariance matrix.
3. Apply matrix algebra (or Excel's matrix functions) to compute the minimum-variance combination of assets.
4. Evaluate the portfolio using a risk-adjusted performance metric — the Sharpe ratio.

The resulting minimum-variance portfolio achieved an expected monthly return of 0.54 % with a standard deviation of 8.03 %, producing a Sharpe ratio of –0.344 when measured against a 3.3% monthly risk-free rate.

This outcome illustrates that even among well-established, large U.S. banks, diversification can reduce volatility; however, when benchmarked against a relatively high risk-free rate, the portfolio's excess return becomes negative. It highlights a key principle of modern portfolio theory: the attractiveness of an investment depends not only on its absolute return, but also on the relative cost of risk.

Small-Bank Portfolio Optimization

The purpose of this section is to demonstrate how to create a minimum-variance portfolio (MVP) using three small U.S. regional banks: Western Alliance (WAL), Zions Bancorporation (ZION), and First Horizon (FHN).

The analysis uses monthly returns from January 2021 through November 2025 to determine how each stock's performance contributes to risk and return, and how diversification can reduce overall volatility.

This report is written in a "teaching" format, so that even readers without a strong math background can follow how the MVP is calculated and interpreted.

Data Overview: Small Banks (Western Alliance, Zions Bancorporation, First Horizon)

The dataset contains monthly percentage returns for each of the three banks, covering 58 months.

A portion of the data is shown below:

Date	WAL	ZION	FHN
Jan-25	0.0518	0.0665	0.0951
Dec-24	–0.1038	–0.0972	–0.0468
Nov-24	0.1249	0.1624	0.2194
...	...	...	...

This is shown by calculating the returns based on the adjusting closing prices from the bank data

we got from Yahoo Finance. The formula for returns is Rt = Pt - Pt-1/Pt-1. What this essentially means is when we look at the data in Yahoo Finance, we are looking at the starting date of one month then subtracting that adjusted closing price by the prior month, and then dividing that by the prior months closing price. For example, if we look at the data in Yahoo Finance for adjusting closing price, we see that Jan 1-25 has an adjusted closing price of $86.23, where as the month prior Dec 1-24 has a price of $81.98. With this information our formula would look like this: 86.23 – 81.98/81.98 = 0.0518. This is what we get for Jan-25 Rate of Return. I'm not going to show all of these calculation because that would be long and tedious, not to mention boring, but now you know how the rate of returns were calculated here.

Step 1: Calculate Average Return and Risk for Each Bank

To begin, we find the mean (average) monthly return and standard deviation for each stock.

Bank	Mean (Average %)	Std. Deviation (%)
WAL	1.68 %	14.32 %
ZION	1.19 %	11.30 %
FHN	1.84 %	11.14 %

Finding this is just basic statistics, we find the mean by adding up all the return values we got for each bank, we add them altogether, and then divide by the number of values where in this case we have 58 values so we divide that big sum by 58 and that's how we get our mean for each bank. Standard deviation has a more complex formula that almost no one uses because its long, complicated, and in many ways impractical. I just used the function in excel "STDEV.S" and then highlighted all the returns we had for each bank and calculated the standard deviation for each bank.

These figures show that while FHN had the highest average return, WAL was the most volatile (its returns fluctuated the most).

Step 2: Examine How Their Movements Relate (Covariance)

To see how the banks move together — sometimes up, sometimes in opposite directions — we create a covariance matrix. This tells us how much the returns of two banks tend to change in the same direction.

	WAL	ZION	FHN
WAL	0.020165	0.014073	0.009391
ZION	0.014073	0.012560	0.008247
FHN	0.009391	0.008247	0.012198

These are monthly covariances (the units are "squared returns").

A positive number means the pair tends to move in the same direction.

This is another example like standard deviation where the formula is more complex, I just did the excel function "Covariance.P," except in this case its not as simple as going through all the values of each individual bank and calculating it for each, Covariance requires two sets of data. In this case we're going to highlight all of the returns for WAL and then highlight it again for our second set of data, so we get covariance of WAL with just its own data going into itself, I just call this 1:1. Then we do that for ZION and FHN which gets us 2:2 and 3:3, and then we combine the data sets in covariance like taking WAL's returns and putting it together with ZION and FHN's data I the covariance formula, there we get 1:2 and 1:3, and then ZION and FHN go into covariance formula which gets us 2:3. This may seem strange how I worded this but its how these values are placed in our handmade covariance matrix. When we make the matrix we start with our first row with three values, those three values will be whatever we get for 1:1, 1:2, and 1:3. Then we go to the second row and do 2:1, 2:2, and 2:3, and same for the third row. We only have three banks so this will be a 3x3 matrix. Notice also how some of the values repeat, that's because some these values are essentially the same calculations, like 1:3 is the same as 3:1 because its using the same sets of data just in a different order. The only unique values are the symmetric ones like 1:1, 2:2, and 3:3

Step 3: Finding the Minimum-Variance Portfolio

The goal is to find the mix of the three banks (their weights) that gives the portfolio the lowest possible risk.

Mathematically, this involves computing:

$$\mathbf{w} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^{\top}\Sigma^{-1}\mathbf{1}}$$

where

- Σ^-1 = the inverted covariance matrix

- 1 = a column of ones (representing each stock)

Inverted covariance matrix

	WAL	ZION	FHN
WAL	227.542	−251.619	−5.058
ZION	−251.619	421.414	−91.203
FHN	−5.058	−91.203	147.532

We got these values by using the "Minverse" function on excel with our covariance matrix that we just made and it gives us this inverted matrix

Multiply by column of ones

This part requires matrix multiplication. By this point we should also have a 1x3 matrix that all just says 1 because these values will end up equaling 1. We have a 3x3 inverted matrix, and 1x3 1 matrix, how this multiplication works is we multiply the three values in our first row of our 3x3 matrix and multiply it downwards from the 1 matrix. After we get those three values we add them together, and that gets us our first out of three values we should end up getting for this matrix. In this case the first value will look like this (227.542)(1) + (-251.619)(1) + (-5.058)(1) = -29.134. This gets us our first value for our new 1x3 matrix, and using the same process we should get -29.134, 78.593, and 51.271

Normalize so that weights sum to 1

Sum of values = (–29.134 + 78.593 + 51.271) = 100.730

Each weight = that value ÷ 100.730

When we've done this, this essentially translated into what our denominator for the weights formula is. With this we would do -29.134/100.730, 78.593/100.730, and 51.271/100.730, which get us our new weights of -0.2892277, 0.78023075, and 0.50899691, and all of these add up to 1 as well which satisfies the portfolio condition. These will be used to calculate our expected return and our portfolio variance and standard deviation

New Weights: -0.2892277, 0.78023075, and 0.50899691

Interpretation:

- WAL has a small negative position (mathematically offsets risk).

- ZION and FHN make up the majority of the portfolio.

Does this mean we did something wrong if we have a bank that has a negative risk? No. From our model it shows we have a negative value for WAL, which is suggesting that in order to minimize risk, we would have to sell or offset a small amount of WAL instead of owning it. In theory a short position earns a positive return when that stock goes down, which helps balance or hedge the movement of the other two stocks

Step 4: Calculate Expected Portfolio Return

Expected Portfolio Return asks again for matrix multiplication. This time we're doing matrix multiplication with our 1x3 weight matrix (-0.2892277, 0.78023075, 0.50899691), and our 3x1 mean or average matrix (0.01682244, 0.01187195, 0.01841184). In this case when we do our multiplication and add everything together, we should end up getting 0.013768913 or 1.38%

Expected portfolio return = 1.38 % per month

Step 5: Calculate Portfolio Variance (Risk)

Next is to calculate the Portfolio Variance (Risk). This was a little more complicated so I just used excel to find this, and the formula looked like this

=MMULT(TRANSPOSE(weights), MMULT(covariance_matrix, weights))

My result was 0.009927465

Then in order to find our portfolio standard deviation we just take the variance that we just found and square root it which ends up getting 0.099636663 or 9.96%

Portfolio standard deviation = 9.96 % per month

Step 6: Sharpe Ratio – Risk-Adjusted Performance

To compare returns relative to risk, we calculate the Sharpe Ratio.

Formula:

S = Expected returns – risk free rate/portfolio standard deviation

We already calculated expected returns and portfolio standard deviation which was 0.01377 and 0.09964, but what is risk free rate? This is a rate given at certain times to banks. As of November 14, 2025, the 3-month treasury bill yield (risk free rate) was 3.95%, and since this is in a year, there would be approximately 0.33% monthly. From this our formula would look like

S = 0.01377 – 0.0033/0.09964 = 0.105

Sharpe Ratio = 0.105 per month, or about 0.36 annualized

($0.105 \times \sqrt{12}$)

Interpretation:

For every 1 unit of risk, this portfolio earns about 0.105 units of return above the risk-free rate.

Discussion and Teaching Notes

- This analysis shows the power of diversification: by combining three correlated but not perfectly correlated bank stocks, total risk dropped to 9.96 % per month — lower than any single stock's volatility.
- The negative weight for WAL doesn't mean borrowing shares — it simply indicates that WAL's movements offset some of the portfolio's total variability.
- The Sharpe ratio provides a risk-adjusted view — showing efficiency rather than raw return.

Put simply:

Individually, each bank's stock bounces around a lot. Together, they form a smoother-riding portfolio with roughly the same return potential.

Limitations and Next Steps

- The analysis assumes past performance predicts future risk relationships, which is not guaranteed.
- Only three banks were included; adding broader financial institutions could yield even better diversification.
- The risk-free rate was based on November 2025 yields — future studies could use time-varying or average rates for more accuracy.

Conclusion

Through this exercise, we demonstrated how to systematically:

1. Measure the average returns and risks of individual stocks.
2. Understand how they move together using covariance.
3. Use matrix algebra (or Excel functions) to calculate the mix that minimizes overall risk.

4. Evaluate the results using a risk-adjusted metric (the Sharpe ratio).

The final minimum-variance portfolio achieved an expected monthly return of 1.38 % with a risk level of 9.96 %, producing a Sharpe ratio of 0.105.

This approach shows that even within a single industry (small U.S. banks), combining assets intelligently can significantly stabilize returns — a central lesson in modern investing.

V. **Interpretation and Comparisons**

So we've looked at three groups of banks, successful banks, some regular banks, and some small banks, and we've used Markowitz model to calculate minimum variance portfolio and what investors can do to minimize risk, but what exactly does all the values we've calculated here mean? Lets look into that

1. Return Patterns Across Groups

- Successful Banks offered the highest expected return (2.22 %), consistent with their strong earnings, scale, and market resilience.

- Regular Banks posted the lowest return (0.54 %), reflecting more conservative operations and lower growth potential.
- Small Banks fell in the middle with 1.38 %, balancing moderate growth and higher exposure to local-economic factors.

In other words, moving from smaller to larger institutions generally increases average return potential up to the most successful tier, though not proportionally to risk.

2. Volatility and Risk Comparison

- Regular Banks had the lowest volatility (8.03 %), showing stable profit streams and consistent performance.
- Successful Banks experienced a slightly higher volatility (6.85 %), yet their better return profile indicates that some of that risk translated into productive reward.
- Small Banks were the most volatile (9.96 %) — typical for institutions more sensitive to local shocks and lower liquidity.

So in relative terms, "stable but slow" describes regular banks, while small banks are "risky but occasionally rewarding."

3. Risk-Adjusted Performance

The Sharpe ratio compares excess return per unit of risk after subtracting the 3.3 % risk-free benchmark.

- Small Banks: 0.105 → small positive reward for risk taken.
- Regular Banks: –0.344 → return not sufficient to beat the risk-free rate.

- Successful Banks: –0.158 → closer to breakeven but still below the high benchmark.

It's crucial to note that the chosen risk-free rate (3.3 % monthly ≈ 47 % annualized) is unusually high for most economic environments. Using that gauge exaggerates the negative Sharpe scores for Regular and Successful Banks; with a typical 0.3 % monthly risk-free rate, both would become solidly positive.

Still, within a consistent framework, Small Banks had the best relative Sharpe ratio, meaning their expected excess return compensated slightly better for their higher volatility.

4. Economic Interpretation

- Small Banks: Their diversification among highly idiosyncratic regional institutions limited extreme losses and yielded modest excess gains despite higher volatility.
- Regular Banks: The middle tier's weaker Sharpe stems from defensive positioning — less volatility, but also minimal expected excess return, leaving little margin above the risk-free rate.
- Successful Banks: Strong absolute performance (high returns, low risk) yet under the high benchmark, resulting in a mildly negative Sharpe. This indicates that while they produce stable growth, their incremental reward for risk is diluted by market efficiency at the top.

5. Key Insights

1. Size and quality do not guarantee superior risk-adjusted returns. Regular and Successful Banks are safer in absolute volatility terms, but when accounting for the risk-free benchmark, their reward per unit of risk diminishes.
2. Diversification benefits persist even within a single industry. Each MVP achieved lower risk than the individual stocks composing it.
3. The Small-Bank portfolio demonstrated the most efficient trade-off between risk and return under the uniform risk-free assumption.
4. Successful Banks lead in raw performance — investors seeking higher total return might favor them despite a less favorable Sharpe under this high-rate environment.
5. Regular Banks represent stability but limited upside, explaining their negative Sharpe despite consistent outcomes.

6. Concluding Remarks

Comparing all three groups reinforces a central lesson of modern portfolio theory:

> Performance must always be judged relative to both volatility and the prevailing risk-free opportunity.

Even within a single industry, the composition of banks—small, regular, or successful—changes the efficient frontier of risk and return.

Each group's portfolio highlights how institution size, market scope, and diversification interact to define investment efficiency:

- Small Banks: Higher volatility, modest excess return → risk-accepting investors.

- Regular Banks: Low volatility, low return → capital-preserving investors.
- Successful Banks: High return, low risk → growth investors seeking stability.

Collectively, these comparisons show how sophisticated allocation—even among seemingly similar assets—can tailor performance to investors' unique risk preferences.

VI. **Actual Results**

Main Findings

Based on historical data analysis and the application of mean-variance optimization, we expect the research to reveal portfolio allocations that minimize total risk for each group of financial institutions. The resulting efficient frontier should clearly illustrate the fundamental investment trade-off: achieving higher expected returns generally requires accepting higher levels of risk.

Preliminary findings from the small-, regular-, and successful-bank subsets already demonstrate this principle. Each group forms its own point along the broader frontier, with small banks exhibiting higher volatility but slightly better risk-adjusted efficiency, and successful banks achieving the highest absolute returns with the lowest total risk. These patterns suggest that even within a single industry, diversification and institution size produce measurable differences in portfolio behavior.

Examples & Applications

By constructing and testing optimized portfolios across three tiers of U.S. banks, we can later extend the same process to a combined dataset that includes all groups. This combined analysis is expected to show how diversification across bank types further smooths risk while potentially enhancing returns — an empirical demonstration of modern portfolio theory's benefits at the industry level.

Visualizations

The updated analysis will continue to employ efficient-frontier graphs and covariance heatmaps. The frontier plots will make it visually clear how each bank group occupies a distinct risk-return region, while the heatmaps reveal the degree of co-movement among assets within and across those groups.

VII. **Conclusion & Future Work**

Summary

This project uses linear algebra and statistical tools to optimize portfolio allocation, creating a mathematical framework for balancing risk and return. The project aims to demonstrate the practical use of mathematical modeling in finance and how quantitative methods can aid in risk management.

Implications

The project will show how covariance matrices and optimization techniques are used in portfolio management. It also provides tools and frameworks that can be used for further research or in professional settings.

Potential Contributions

This research makes a significant contribution by offering a clear, practical demonstration of how fundamental concepts such as covariance matrices and advanced optimization techniques are actively applied within the realm of portfolio management. Beyond theoretical insights, it also provides valuable tools and frameworks that are designed to be adaptable, enabling future research endeavors and offering practical utility for professionals in the financial industry.

Next Steps

Future work includes incorporating real-world constraints like transaction costs and leverage limits to make the model more practical. Another direction is to explore machine learning techniques for portfolio optimization, which could improve the model's accuracy and adaptability.

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